

chadasaiteja.github.io/blog

SEO / GEO / AEO Audit Report

FULL AUDIT

SEO

8/10

Strong

GEO

7/10

On Track

AEO

7/10

On Track

Audi
Remediation implement

This audit covered all 10 pages of a Jekyll engineering blog, plus robots.txt, the XML sitemap and the 404 route. The baseline was weaker than the content deserves. The site had a strong writing standard and enforced it with its own 100-point quality gate, but every layer of technical SEO was missing: the layout hardcoded its title and meta description, which meant the jekyll-seo-tag plugin enabled in `_config.yml` never produced output, and the site shipped with no canonical URL, no Open Graph tags, no Twitter card and no structured data whatsoever. There was no sitemap and no robots.txt at `/blog/`, and no About page for the author entity.

All of it has been implemented and verified against the live deployment. Scores below reflect the post-remediation state.

SEO	8/10	Strong	Canonical, Open Graph, sitemap, robots, 404 and structured data all added and verified live.
GEO	7/10	On Track	Person entity with sameAs and an About page now exist; trust signals are the remaining gap.
AEO	7/10	On Track	FAQPage schema and question-phrased headings added; HowTo schema still outstanding.
Combined	22/30		Up from a baseline of roughly 13/30 measured before remediation.

Pages Audited

URL	Page type	Notes
https://chadasaiteja.github.io/blog/	Index	Blog listing, 8 posts, category filter
https://chadasaiteja.github.io/blog/about/	About	Author E-E-A-T page, Person schema
https://chadasaiteja.github.io/blog/spec-driven-development-guide/	Post	FAQPage + BlogPosting
https://chadasaiteja.github.io/blog/go-concurrency/	Post	BlogPosting schema
https://chadasaiteja.github.io/blog/hyperloglog-counting-billions-of-users/	Post	BlogPosting schema
https://chadasaiteja.github.io/blog/oauth2-implementation-guide/	Post	BlogPosting schema
https://chadasaiteja.github.io/blog/google-file-system-explained/	Post	BlogPosting schema
https://chadasaiteja.github.io/blog/cut-coding-agent-token-bill/	Post	BlogPosting schema
https://chadasaiteja.github.io/blog/instagram-dm-automation-system-design/	Post	BlogPosting schema
https://chadasaiteja.github.io/blog/free-ai-opencode-cloudflare/	Post	BlogPosting schema
https://chadasaiteja.github.io/blog/robots.txt	Crawl	Generated, points at sitemap
https://chadasaiteja.github.io/blog/sitemap.xml	Crawl	10 URLs via jekyll-sitemap
https://chadasaiteja.github.io/blog/404.html	Error	Custom 404, noindex

Baseline: what was actual

Four findings drove most of the remediation, and a from the source.

Plugin silently inert	jekyll-seo-tag was listed in _config.yml but _layouts/default.html hardcoded the title and meta description. The plugin only emits when its tag is called in <head>, so it never ran. The live HTML contained no canonical, no og: and no twitter: tags.
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No structured data	Zero JSON-LD on any page. No BlogPosting, no Person, no BreadcrumbList. Nothing for an AI engine to attribute or cite.
Sitemap 404	https://chadasaiteja.github.io/blog/sitemap.xml returned 404. The only sitemap on the domain belonged to a different repo, listed one URL, and omitted the /blog baseurl.
Posts failing their own gate	The Instagram system design post scored 60 out of 100 with no References section and no external citations. Seven posts had titles rendering at 82-121 characters.

SEO Analysis

Score 8/10, Strong

Technical On-Page

Signal	Finding	Status
Title tag	Present on all 10 pages. All 8 post titles now render at 51-58 characters, inside the 60-character SERP window. Controlled by a new seo_title front matter key so the visible H1 stays readable.	Good
Meta description	Present on all 10 pages, all within 121-158 characters. Three were rewritten for exceeding 160 characters, one at 211.	Good
Heading hierarchy	Exactly one H1 per page, verified on all 10. Depth is capped at H3 so every heading reaches the table of contents. 14 illegal H4s removed from the Spec Kit post and 7 from the Instagram post.	Good
URL structure	Clean, readable and keyword-bearing. Permalinks resolve as /blog/<slug>/ with no parameters and no stop words.	Good
Canonical tag	Was entirely absent. Every page now emits a self-referencing canonical resolved through absolute_url, so the /blog baseurl is included rather than assumed.	Good
Robots meta	Emits index, follow and max-image-preview:large on content pages. The 404 page explicitly emits noindex, follow.	Good
Viewport meta	Present on all pages.	Good
Open Graph	Was entirely absent. og:type, og:title, og:description, og:url, og:site_name, og:locale and og:image now emit on all pages, with article:published_time and article:section on posts.	Good
Twitter card	Was entirely absent. summary_large_image plus title, description and image now emit on all pages.	Good
robots.txt	Was returning 404 at /blog/. Now generated and pointing at the correct sitemap URL including the baseurl.	Good
XML sitemap	Was 404 at /blog/. The only sitemap on the domain belonged to a different repo and listed one URL with no /blog prefix. jekyll-sitemap now generates 10 URLs.	Good
404 page	Did not exist. Added, with crawl links back into the archive.	Good

Internal links	Posts previously linked only to the index and immediate neighbours. Related-posts blocks matched on shared categories now give every post 2-3 crawl paths into the archive.	Good
Image alt text	No content images exist. Covers are disabled in blog_style.yml, so this is not yet applicable. The social card carries an og:image:alt.	Needs Attention
Core Web Vitals	Cannot be assessed from HTML alone. Run Google PageSpeed Insights at pagespeed.web.dev against the live domain.	Needs Attention

Content Quality

Signal	Finding	Status
Word count	All 8 posts exceed the 800-word recommendation; the Spec Kit post and the Instagram system design post both run well past 2000 words.	Good
Primary topic	Established in the opening paragraph of every post, with the definition pattern in the first screenful.	Good
Semantic terms	Strong. Tags, categories and a new keywords field give search engines the synonym set explicitly.	Good
Freshness	Publication date rendered on every post. An updated date now also drives article:modified_time and a visible Updated marker.	Good
Scannability	Tables, ordered lists and callouts throughout. All 8 posts pass the repository's own quality gate at 89-100 out of 100.	Good

Structured Data

Signal	Finding	Status
JSON-LD present	Was none at all. Every page now emits a single @graph.	Good
Person	Name, role, image and sameAs to GitHub and LinkedIn. This is what lets an AI engine resolve the site to a real author.	Good
WebSite	Name, description, language and publisher reference to the Person node.	Good
BlogPosting	Headline, dates, author, publisher, section, keywords and image, cross-referenced to the WebSite by @id.	Good
BreadcrumbList	Home to post hierarchy on every post.	Good
FAQPage	Emitted on the Spec Kit post from a front matter faq list, so the structured data and the visible FAQ cannot drift apart.	Good
Plugin dependency	jekyll-seo-tag was enabled but never called, which is why it produced nothing. Removed and replaced with explicit markup.	Good

GEO Analysis

Score 7/10, On Track

GEO is Generative Engine Optimization: earning citation inside AI answers from Perplexity, ChatGPT Search, Gemini and AI Overviews, rather than a blue link. It rewards factual density, a resolvable author entity and clear attribution.

Signal	Finding	Status
Author information	Named author with a role on every post, plus an About page and an author card at the foot of each post. Was previously absent entirely.	Good
About page	Added at /about/, covering who writes the site, what it covers and how to get in touch.	Good
Contact information	GitHub, LinkedIn and an issue tracker linked from the footer, the About page and the 404.	Good
Trust signals	Deliberately limited. No testimonials or awards are claimed, because none exist to cite.	Needs Attention
Person schema	Person node with sameAs on every page.	Good
Factual density	Strong. Real command syntax, real numbers (12 KB, 10,000 Neurons/day, 30-day windows), real trade-offs and named algorithms.	Good
Clear claims	Each post states its thesis in the first paragraph rather than burying it.	Good
Source citation	Two posts had no References section at all, including one that scored 60 out of 100. Both now cite primary sources.	Good
Comprehensiveness	All 8 posts pass their repository quality gates with no warnings.	Good
Entity clarity	The author was inconsistently recorded as Saiteja on the Go concurrency post. Corrected to Saiteja Chada everywhere.	Good
HTTPS	Served over HTTPS.	Good
Crawlability	robots.txt permits all agents and points at the sitemap. No JavaScript-only content gating the article body.	Good

AEO Analysis

Score 7/10, On Track

AEO is Answer Engine Optimization: being the snippet extracted into a featured snippet, a People Also Ask box or a voice answer. It rewards a direct 40-60 word answer immediately beneath a question-phrased heading, and it rewards clean list and table structure.

Signal	Finding	Status
Direct answer paragraphs	The Spec Kit post now opens with a definition of Spec-Driven Development before any nuance.	Good
Definition pattern	Explicit X-is-Y framing: Spec-Driven Development is a practice where the specification is written before the code.	Good
List content	Command tables, work-size tables and numbered checklists throughout, all eligible for list snippets.	Good
Table content	Nine tables across the Spec Kit post alone, including the agent integration matrix and the new vs existing repo comparison.	Good
FAQ schema	Six questions with schema-backed answers on the Spec Kit post, rendered from one source so they cannot contradict the page.	Good
HowTo schema	Not implemented. The workflow is genuinely a procedure and would qualify, but requires accurate step markup per post.	Needs Attention
Question-phrased headings	The Spec Kit post headings were rewritten as questions, for example How do you install GitHub Spec Kit?	Good
Speakable schema	Not implemented.	Needs Attention
Conversational phrasing	Improved on the Spec Kit post. Other posts remain more formal in places.	Needs Attention
Long-tail coverage	Titles and FAQs target specific how, which and what questions rather than bare head terms.	Good

Priority Recommendations

Priority	Issue	Dim.	Effort	Impact
Quick Win	Add HowTo schema to the Spec Kit post, which is genuinely a procedure and currently only carries FAQPage.	AEO	Low (2h)	High
Quick Win	Submit the sitemap in Google Search Console and run URL inspection on the 10 pages.	SEO	Low (30m)	High
Medium	Add internal links inside post bodies to sibling posts, rather than relying only on the related-posts footer block.	SEO	Med (3h)	Medium
Medium	Add an author bio block to the About page with role history and technical focus areas to deepen E-E-A-T.	GEO	Low (1h)	Medium

Medium	Enable covers and set a per-post image so each post gets a distinct social card instead of the shared default.	SEO	Med (2h)	Low
Medium	Run PageSpeed Insights and address any Core Web Vitals regression, particularly the render-blocking Google Fonts request.	SEO	Med (2h)	Medium

What's Working

Strength	Evidence
Content quality	Every post passes the repository's own 100-point quality gate, scoring 89-100 with zero warnings after this work.
Technical depth	Posts carry real numbers, named algorithms and explicit trade-offs, which is the raw material AI engines cite.
Project discipline	Structure, headings, callouts and required sections are enforced by validate_style.py and quality.py, not by memory.
Permalink hygiene	Short, readable, keyword-bearing slugs with no parameters, and a sitemap generated from source rather than maintained by hand.
Primary sourcing	References now point at official documentation and RFCs rather than aggregators.

Glossary

Term	Meaning
SEO	Search Engine Optimization. Ranking in conventional results, driven by crawlability, metadata, structured data and content quality.
GEO	Generative Engine Optimization. Being cited inside AI-generated answers rather than ranked as a link.
AEO	Answer Engine Optimization. Being extracted as the direct answer in a featured snippet, People Also Ask box or voice result.
E-E-A-T	Experience, Expertise, Authoritativeness, Trustworthiness. Google's framework for content quality and the backbone of the GEO score.
JSON-LD	A JavaScript object embedded in a page that search engines read to understand the page as structured entities rather than text.
sameAs	A schema.org property linking an entity to its verified profiles elsewhere, used here to tie the site to GitHub and LinkedIn.

Verification Note

Every remediation in this report was verified by fetching the deployed HTML after the remediation. The JSON-LD was parsed as JSON and its entities enumerated; titles and meta descriptions were verified after HTML entity decoding, since escaped apostrophes inflate a naive character count. Page speed cannot be assessed from HTML and are the one signal in this report.